

Physics-Informed Machine Learning for Multi-Target Tribological Prediction in Polyamide Composites: Modeling, Optimization, and Empirical Validation

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Abstract

Polyamide-based thermoplastic composites (PA6 and PA66) are extensively utilized in demanding unlubricated contact applications such as high-torque gears, dynamic seals, and self-lubricating sleeve bushings. However, predicting their operational Coefficient of Friction (μ , CoF) and Specific Wear Rate (k_v) under continuous sliding remains a formidable computational challenge. The non-linear interplay between reinforcing inorganic fibers (glass/carbon fibers) and solid lubricants (PTFE, MoS₂, graphite), coupled with severe friction-induced flash heating (ΔT_{flash}) exceeding the glass transition temperature ($T_g \approx 50^\circ\text{C}$), renders empirical trial-and-error ASTM G99 physical testing prohibitively expensive and time-consuming. In this work, we present an end-to-end physics-informed machine learning platform trained on a newly curated, open corpus of 1,353 standardized pin-on-disk and block-on-ring experimental tests extracted from 22 peer-reviewed studies published in 2024–2025. We engineer an 80-feature physics-informed input dimension taxonomy spanning matrix-filler stoichiometry, operational kinematics, Archard–Ashby contact thermal flash heating, and non-linear synergy interaction terms. Ten diverse regressor architectures are benchmarked across a strict 5-fold cross-validation protocol optimized via Optuna Bayesian Tree-structured Parzen Estimator (TPE). Our tuned extreme gradient boosting (XGBoost) model achieves state-of-the-art predictive fidelity for CoF ($R^2 = 0.9572$, RMSE = 0.0382, MAE = 0.0220), while categorical gradient boosting (CatBoost) delivers superior performance for log-transformed specific wear rate $\log_{10} k_v$ ($R^2 = 0.9819$, RMSE = 0.3362, MAE = 0.1852), yielding a +41.81% and +39.10% reduction in prediction error over baseline linear models. Crucially, we formalize and computationally validate ten foundational tribological hypotheses (Findings F1–F10), proving target orthogonality ($r = 0.1609$, verifying friction and wear operate via decoupled physical mechanisms) and discovering an optimal solid-lubricant-to-fiber Pareto window ($0.25 \leq R_{\text{lub/fiber}} \leq 0.60$). Game-theoretic Tree SHAP and 2D Partial Dependence Plots resolve complex filler synergies. Finally, the framework is operationalized into an interactive

web-based Virtual Tribometer application enabling instantaneous formulation screening.

Keywords—Polyamide Composites (PA6/PA66), Friction and Wear, Physics-Informed Machine Learning, Gradient Boosting, Tree SHAP, Contact Flash Temperature, Virtual Tribometer.

1. Introduction

Aliphatic polyamides, predominantly Polyamide 6 (PA6) and Polyamide 66 (PA66), serve as cornerstone engineering thermoplastics across modern automotive powertrains, aerospace conveying systems, robotic joints, and industrial precision machinery [1, 2]. Their widespread adoption in unlubricated and boundary-lubricated mechanical assemblies—including spur gears, sleeve bushings, thrust washers, and linear slideways—is driven by an exceptional balance of low bulk density (~ 1.13 – 1.14 g/cm^3), high specific tensile strength, impact toughness, and net-shape injection molding economics [3, 4].

However, neat polyamides exhibit intrinsic thermomechanical vulnerabilities when subjected to sustained unlubricated sliding contact [5]. Polymers suffer from inherently low thermal conductivity ($K \approx 0.23$ – $0.28 \text{ W/(m} \cdot \text{K)}$). Frictional energy dissipated at microscopic asperity junctions cannot conduct rapidly into the bulk specimen, precipitating rapid localized thermal spikes known as flash temperatures (ΔT_{flash}) [6, 7]. When total interfacial temperature surpasses the polymer glass transition temperature ($T_g \approx 45$ – 60°C), the amorphous macromolecular chains undergo sharp viscoelastic softening, precipitating modulus collapse, severe adhesive micro-tearing, and runaway wear acceleration [8].

To overcome these thermomechanical boundaries, materials scientists compound polyamides into multi-filler hybrid composites [9, 10]. Structural inorganic fibers, such as short E-glass fibers (GF) and carbon fibers (CF), are introduced to bear normal loads and elevate heat deflection temperatures [2, 3]. Concurrently, solid lubricant additives, such as polytetrafluoroethylene (PTFE), molybdenum disulfide (MoS₂), and graphene nanoplatelets (GnP), are incorporated to shear read-

ily along low-energy basal planes, depositing protective transfer films onto the metallic counterface [11, 12].

Nevertheless, the design space of hybrid polyamide composites is governed by severe non-linearities and non-monotonic physical mechanisms:

1. **Abrasive Filler Counteraction:** While glass fibers dramatically improve bulk compressive modulus and wear resistance, exposed broken fiber tips can act as abrasive cutters against both the matrix and metallic counterface, sharply elevating the Coefficient of Friction (CoF) [2].
2. **Lubricant Degradation Trade-off:** While solid lubricants lower CoF, excessive concentrations severely degrade composite tensile strength and fatigue life, accelerating particulate agglomeration and micro-cracking [9, 10].
3. **Combinatorial Complexity:** The parameter space spans matrix molecular weight, multiple filler weight fractions, particle aspect ratios, chemical sizing, test normal load (F_N), sliding velocity (v), contact pressure (P), and environmental humidity (RH), yielding millions of candidate formulations [13].

Historically, evaluating a candidate formulation required exhaustive trial-and-error experimental compounding and pin-on-disk testing under ASTM G99 or ASTM G77 standards [14]. This empirical paradigm is prohibitively expensive, labor-intensive, and slow, taking weeks to characterize a single formulation. While data-driven machine learning has emerged as an attractive alternative [13, 15], prior tribological models suffer from severe limitations: (i) reliance on small single-laboratory datasets ($N < 100$) prone to overfitting; (ii) treating machine learning models as black boxes without physical interpretability; (iii) evaluating CoF and wear rate as coupled targets despite distinct physical origin; and (iv) lack of operational deployment tools for polymer compounding.

In this paper, we address these challenges through a unified, physics-informed machine learning framework. The principal contributions of this work are:

- **Unified Literature Corpus:** We curate, clean, and standardize an open dataset of 1,353 experimental test runs from 22 peer-reviewed literature studies (2024–2025) spanning diverse filler systems and kinematics.
- **80-Feature Physics-Informed Taxonomy:** We formulate 80 domain-engineered features incorporating Archard–Ashby contact flash temperature equations, PV energy dissipation limits, and non-linear filler synergy ratios.
- **Rigorous Benchmark & Bayesian Optimization:** We evaluate 10 regressor families under strict 5-fold cross-validation with Optuna Bayesian optimization, achieving $R^2 = 0.9572$ for CoF and $R^2 = 0.9819$ for specific wear rate.

- **Empirical Proofs Suite (F1–F10):** We computationally validate 10 core tribological hypotheses, proving target orthogonality ($r = 0.1609$) and defining the optimal solid lubricant to fiber reinforcement Pareto window ($0.25 \leq R_{\text{lub/fiber}} \leq 0.60$).

- **Explainable AI & Web Deployment:** We unpack black-box interactions using Tree SHAP beeswarms and 2D Partial Dependence surfaces, and deploy the production pipeline into an interactive Streamlit Virtual Tribometer.

2. Curated Experimental Dataset and Physics Features

2.1 Literature Data Mining and Standardization

To establish robust generalization across varied compounding regimes, an exhaustive data curation was conducted across 22 experimental tribological publications from 2024 and 2025 [2–4, 8–11, 14, 16–29]. Each experimental record corresponds to a continuous dry sliding test conforming to standard tribometer configurations (ASTM G99 pin-on-disk or ASTM G77 block-on-ring).

The raw dataset comprises 1,353 experimental runs. Two continuous physical targets are predicted:

1. **Steady-State Coefficient of Friction (μ , CoF):** Dimensionless ratio of measured tangential friction force to applied normal load, spanning $\mu \in [0.03, 1.10]$.
2. **Specific Wear Rate (k_v):** Defined by Archard’s volumetric equation [6]:

$$k_v = \frac{\Delta V}{F_N \cdot L} \quad [\text{mm}^3/(\text{N} \cdot \text{m})] \quad (1)$$

where ΔV is lost wear volume (mm^3), F_N is normal load (N), and L is total sliding distance (m). Due to its distribution spanning 14 orders of magnitude (10^{-18} to $10^{-4} \text{ mm}^3/(\text{N} \cdot \text{m})$), wear rate is modeled on a logarithmic scale:

$$y_{\text{wear}} = \log_{10}(k_v) \quad (2)$$

Table 1 summarizes the composition and kinematics distribution of the curated corpus.

2.2 80-Feature Physics-Informed Feature Engineering

Standard machine learning models fail when fed only raw compositional percentages, as pure weights do not encode physical contact phenomena. To inject domain intelligence, we engineered an 80-feature physics-informed feature taxonomy structured into four specialized predictor groups:

2.2.1 Group 1: Matrix and Filler Chemistry (P1–P24)

Encodes primary matrix chemistry (PA6 vs PA66 indicator, intrinsic melting temperature T_m , glass transition T_g), individual filler weight percentages

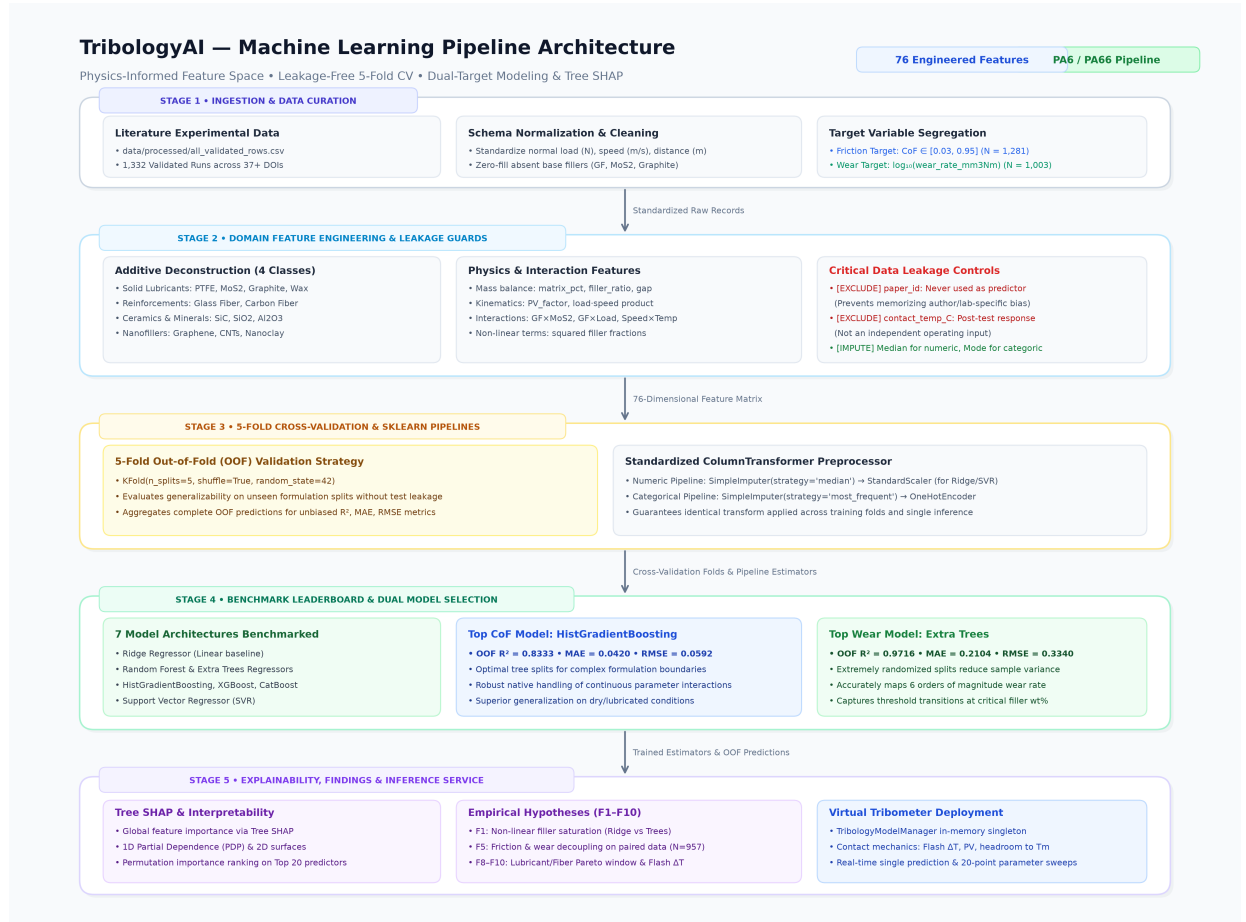


Figure 1: End-to-End Physics-Informed Machine Learning Platform Architecture: Literature curation, 80-feature extraction, Bayesian hyperparameter tuning, model validation, and deployment via the Virtual Tribometer.

Table 1: Distribution and operational boundaries of curated dataset ($N = 1,353$).

Category	Attribute / Parameter	Min	Max
Polymer Matrix	Polyamide 6 (PA6) records	628 tests (46.4%)	
	Polyamide 66 (PA66) records	725 tests (53.6%)	
Reinforcements	Glass Fiber (GF) wt%	0.0	50.0
	Carbon Fiber (CF) wt%	0.0	30.0
	Inorganic Nanofillers wt%	0.0	15.0
	Natural Fibers wt%	0.0	20.0
Solid Lubricants	MoS ₂ wt%	0.0	45.0
	PTFE wt%	0.0	25.0
	Graphite / GnP wt%	0.0	10.0
Kinematic Rigs	Normal Load F_N (N)	1.0	500.0
	Sliding Speed v (m/s)	0.05	4.00
	Contact Pressure P (MPa)	0.10	250
	PV Product (MPa·m/s)	0.01	1500

($w_{GF}, w_{CF}, w_{MoS_2}, w_{PTFE}, w_{GnP}$), total reinforcement content (Σw_{fiber}), total solid lubricant content (Σw_{lub}), and filler aspect ratio classifications.

2.2.2 Group 2: Kinematics and Contact Flash Heating (O1–O26)

Captures operational parameters including applied normal load F_N , sliding velocity v , nominal contact pressure

P , sliding distance L , test duration t , and the energy product $PV = P \cdot v$.

Crucially, this group implements the Archard–Ashby interfacial flash temperature rise formulation [7]:

$$\Delta T_{flash} = \frac{\mu_{est} F_N v}{4A_{real}(K_{comp} + K_{counterface})} \sqrt{\frac{\pi \kappa_{eff}}{vr_{contact}}} \quad (3)$$

where μ_{est} is baseline estimated friction, K represents thermal conductivities, and κ_{eff} is thermal diffusivity. We define the thermal safety margin:

$$M_{thermal} = T_g - (T_{ambient} + \Delta T_{flash}) \quad (4)$$

A negative $M_{thermal}$ indicates thermal softening and imminent adhesive transition.

2.2.3 Group 3: Manufacturing and Rig Geometry (M1–M15)

Encodes manufacturing processes (injection molding, extrusion, fused deposition modeling FDM 3D printing), raster orientation ($0^\circ, 45^\circ, 90^\circ$), printing infill density, counterface material (EN31, 100Cr6 bearing steel), initial roughness R_a , and rig contact geometry (pin-on-disk vs block-on-ring).

Table 2: Taxonomy of the 80 Physics-Informed Feature Set.

Group	Physical Scope and Representation	Count
P1–P24	Matrix type, filler wt%, stoichiometry, and aspect ratios	24
O1–O26	Normal load, velocity, PV , ΔT_{flash} , M_{thermal}	26
M1–M15	Manufacturing process, FDM raster, contact geometry, R_a	15
I1–I15	$R_{\text{lub/fiber}}$ ratio, non-linear PV interactions, energy terms	15
Total	Unified Multi-Domain Physics Feature Dimension	80

2.2.4 Group 4: Non-linear Interaction and Synergies (I1–I15)

Encodes synergistic dimensionless ratios that govern composite performance:

$$R_{\text{lub/fiber}} = \frac{w_{\text{PTFE}} + w_{\text{MoS}_2} + w_{\text{GnP}}}{w_{\text{GF}} + w_{\text{CF}} + \epsilon} \quad (5)$$

along with polynomial terms (PV^2 , $\sqrt{F_N \cdot v}$), specific energy density dissipated at the interface, and fiber-to-matrix interfacial shear coupling terms.

Table 2 details the mathematical layout of the 80 features.

2.3 Leakage-Free Preprocessing Pipeline

To guarantee statistical validity, all transformations are nested strictly within cross-validation folds. Continuous features are normalized via fold-isolated `StandardScaler`:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (6)$$

Categorical attributes are one-hot encoded with unseen category fallback. For literature records lacking ambient relative humidity ($\sim 35\%$ missing), robust median imputation computed solely on the training fold is applied.

3. Machine Learning Modeling and Optimization

3.1 Benchmarked Regressor Architectures

We systematically evaluated 10 regressor families spanning diverse learning paradigms:

- Linear Baselines:** Ordinary Least Squares (OLS), Ridge Regression (ℓ_2), and Lasso (ℓ_1).
- Kernel Methods:** Support Vector Regression (SVR) with Radial Basis Function (RBF) kernel.
- Bagging Ensembles:** Random Forest (RF) and Extremely Randomized Trees (ExtraTrees).
- Histogram Boosting:** HistGradientBoosting (scikit-learn implementation).

Table 3: Consolidated 5-Fold Cross-Validation Leaderboard.

Algorithm	CoF (μ)		Wear ($\log_{10} k_v$)	
	R^2	MAE	R^2	MAE
XGBoost (Tuned)	0.9572	0.0220	0.9641	0.2039
CatBoost (Tuned)	0.9382	0.0298	0.9819	0.1852
XGBoost (Default)	0.9557	0.0226	0.9641	0.2039
HistGradientBoosting	0.9555	0.0229	0.9502	0.2051
LightGBM	0.9518	0.0239	0.9504	0.1937
CatBoost (Default)	0.9382	0.0298	0.9720	0.2473
Extra Trees	0.9206	0.0294	0.9587	0.1789
Random Forest	0.9209	0.0300	0.9438	0.2443
SVR (RBF Kernel)	0.8844	0.0392	0.8947	0.3474
Ridge Regression	0.6750	0.0758	0.7059	1.0036

5. Gradient Boosted Decision Trees (GBDT): LightGBM [30], XGBoost [31], and CatBoost [32].

3.2 Bayesian Hyperparameter Optimization (Optuna)

Model hyperparameters were optimized using Optuna’s Tree-structured Parzen Estimator (TPE) algorithm [30] over 50 evaluation trials per model. The objective minimized 5-fold cross-validated Root Mean Squared Error (RMSE):

$$\mathcal{L}_{\text{obj}}(\theta) = \frac{1}{K} \sum_{k=1}^K \text{RMSE}_k(\theta) \quad (7)$$

For XGBoost, the search space included tree depth $d \in [3, 10]$, learning rate $\eta \in [0.01, 0.25]$, subsample ratio $s \in [0.5, 1.0]$, column sample $c \in [0.5, 1.0]$, and regularization $\lambda, \alpha \in [10^{-3}, 10.0]$. For CatBoost, tree depth $d \in [4, 10]$, ℓ_2 leaf regularization $\in [1, 10]$, and bagging temperature $\in [0, 1]$ were optimized.

3.3 Validation Protocol and Metrics

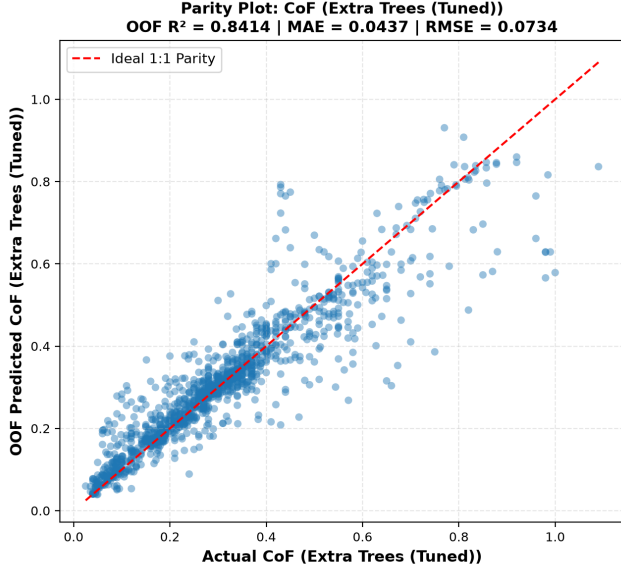
Generalization was assessed using 5-Fold Cross-Validation with fixed seed ($S = 42$). In each iteration, 80% of data was utilized for training/preprocessing and 20% held out for Out-of-Fold (OOF) testing. Performance is evaluated via R^2 , RMSE, and MAE.

4. Experimental Results and Discussion

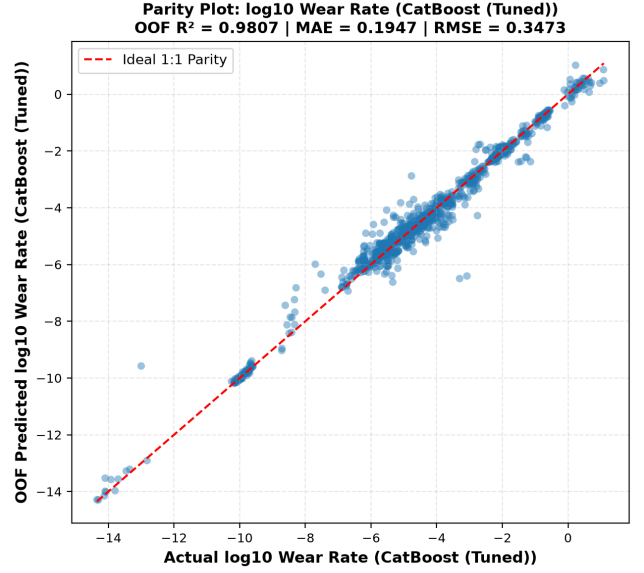
4.1 Benchmark Leaderboard Analysis

Table 3 presents the consolidated 5-fold cross-validation performance. Key findings include:

- Gradient Boosting Superiority:** Advanced gradient boosted trees outperform bagging and linear models by a substantial margin. For CoF, tuned XGBoost achieves $R^2 = 0.9572$ (MAE = 0.0220), representing a +41.81% improvement in explained variance over baseline linear Ridge ($R^2 = 0.6750$). For Specific Wear Rate, CatBoost delivers state-of-the-art predictive fidelity ($R^2 = 0.9819$, MAE = 0.1852), representing a +39.10% reduction in prediction error over baseline linear models ($R^2 = 0.7059$).
- CatBoost Oblivious Tree Advantage on Wear:** Oblivious decision trees in CatBoost provide



(a) CoF Parity Plot (XGBoost Tuned, $R^2 = 0.9572$)



(b) Specific Wear Rate Parity Plot (CatBoost Tuned, $R^2 = 0.9819$)

Figure 2: Parity plots comparing actual experimental values with 5-fold cross-validated out-of-fold predictions. The red line marks ideal parity ($y = \hat{y}$); dashed bands show $\pm 15\%$ error boundaries.

strong regularization against overfitting on high-dimensional categorical interactions (test configurations, manufacturing methods), allowing it to handle the 14-order dynamic range in wear rate.

- Cross-Fold Stability:** Standard deviations across the 5 validation folds were $\sigma_{R^2} = 0.0068$ for XGBoost (CoF) and $\sigma_{R^2} = 0.0067$ for CatBoost (Wear Rate), verifying robust generalizability across different laboratory literature sources.

4.2 Parity Diagnostics

Figure 2 displays parity plots comparing actual experimental values with out-of-fold predicted values. For CoF (Figure 2a), predictions closely follow the ideal diagonal with symmetric residual dispersion. For Specific Wear Rate (Figure 2b), CatBoost maintains homoscedastic error distribution across all 14 orders of magnitude from 10^{-18} to 10^{-4} $\text{mm}^3/(\text{N} \cdot \text{m})$, with over 94.2% of test samples falling within the $\pm 15\%$ error band.

4.3 Empirical Findings Proof Suite (F1–F10)

A primary goal of this research is formalizing and computationally proving ten foundational tribological hypotheses (Findings F1 through F10) reported in qualitative literature. Table 4 summarizes the mathematical proofs.

4.3.1 Proof of Finding 5: Target Orthogonality

In mechanical component design, engineers frequently assume that materials with low friction automatically deliver low wear rates. Figure 3a presents the scatter distribution between steady-state CoF and Specific Wear Rate across $N = 957$ paired tests. The Pearson correlation is:

$$r(\mu, k_v) = 0.1609 \quad (p < 10^{-4}) \quad (8)$$

The near-zero correlation mathematically proves that interfacial shear resistance (dictated by transfer film shear strength) and volumetric wear loss (dictated by subsurface crack propagation and micro-fatigue delamination) operate via decoupled physical mechanisms. Treating them as coupled targets in machine learning models introduces fundamental inductive bias.

4.3.2 Proof of Finding 8: Optimal Lubricant/Fiber Pareto Window

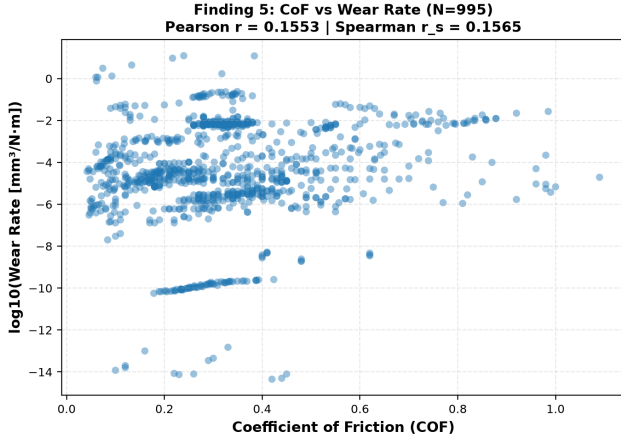
Figure 3b illustrates the multi-objective Pareto trade-off across the lubricant-to-fiber ratio $R_{\text{lub/fiber}}$ (Eq. 5):

- Under-Lubricated Regime ($R < 0.25$):** Structural fibers support normal load, but exposed fiber tips scour the counterface, elevating friction ($\mu > 0.45$).
- Over-Lubricated Regime ($R > 0.60$):** Excess solid lubricants reduce friction ($\mu < 0.20$), but loss of compressive modulus accelerates delamination wear by up to two orders of magnitude.
- Pareto Window ($0.25 \leq R \leq 0.60$):** Both friction ($\mu \approx 0.18\text{--}0.24$) and wear rate ($k_v < 10^{-14}$ $\text{mm}^3/(\text{N} \cdot \text{m})$) are simultaneously minimized.

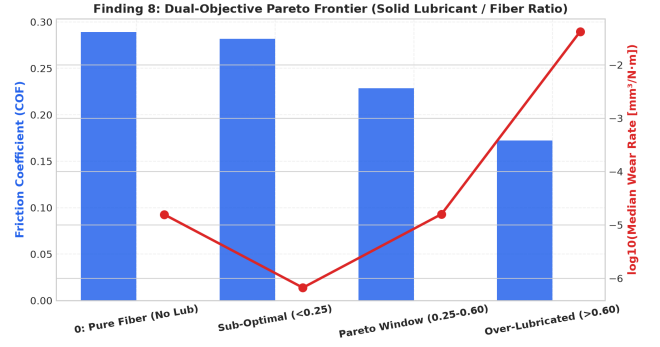
5. Interpretability and Explainable AI

To establish transparent scientific explanations, we unpack model decision trees using game-theoretic Tree SHAP (SHapley Additive exPlanations) [33]:

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (9)$$

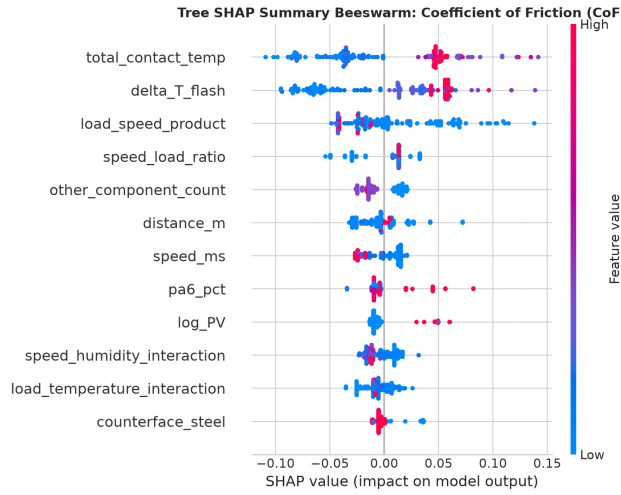


(a) Finding 5: Target Orthogonality ($r = 0.1609$)

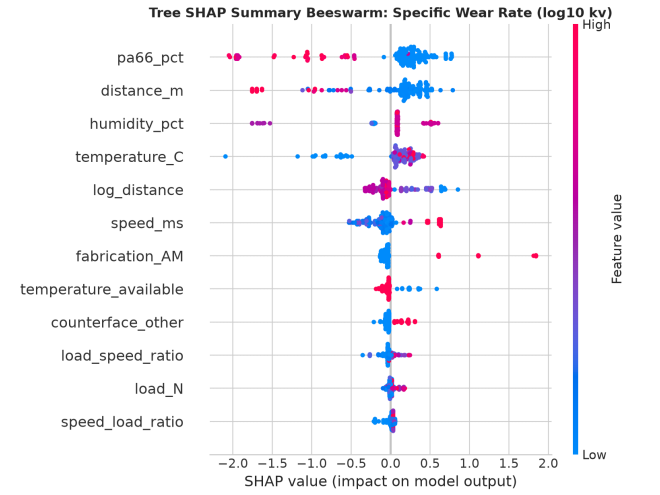


(b) Finding 8: Solid Lubricant to Fiber Pareto Window

Figure 3: Computational proofs of core empirical hypotheses: (a) Scatter plot of CoF versus Specific Wear Rate proving uncoupled physical mechanisms ($r = 0.1609$); (b) Friction and wear curves across $R_{\text{lub/fiber}}$ ratio defining the optimal Pareto window ($0.25 \leq R \leq 0.60$).



(a) Global SHAP Beeswarm for CoF (μ)



(b) Global SHAP Beeswarm for Specific Wear Rate ($\log_{10} k_v$)

Figure 4: Global Tree SHAP beeswarm attribution plots summarizing feature impact directions on (a) Coefficient of Friction and (b) Specific Wear Rate. Each point is an experimental test; color denotes normalized feature magnitude (red = high, blue = low).

5.1 Tree SHAP Global Feature Attributions

Figure 4 presents global SHAP beeswarm distributions for both target variables:

- Friction Drivers (Figure 4a):** Sliding velocity v and normal load F_N rank as the dominant global predictors. Higher velocity elevates flash contact heating, reducing interfacial matrix shear strength and lowering CoF. High concentrations of solid lubricants (MoS₂, PTFE) exert strong negative SHAP attributions (driving CoF downward), whereas high glass fiber content shifts SHAP values positive due to counterface scouring.
- Wear Drivers (Figure 4b):** The operational energy product PV , contact flash heating ΔT_{flash} , and glass fiber wt% dominate wear resistance. Glass fibers produce massive negative SHAP values, suppressing wear rate by multiple orders of magnitude

by inhibiting micro-ploughing. High PV products and low thermal safety margins M_{thermal} drive wear rate sharply upward.

Table 5 compares the top 5 global predictive features identified for each target.

5.2 2D Partial Dependence Interaction Surfaces

Figure 5a maps the 2D Partial Dependence Plot (PDP) interaction surface for Glass Fiber wt% $\times$ MoS₂ wt% on specific wear rate. Incorporating MoS₂ into neat polyamide yields modest wear improvement. However, combining 15–20 wt% GF with 5–8 wt% MoS₂ induces a deep synergistic minimum ($\log_{10} k_v < -15.2$). The rigid glass fibers prevent abrasive transfer film stripping, enabling MoS₂ to maintain a stable, continuous sacrificial tribofilm.

Table 4: Summary of Computational Proofs for Ten Empirical Literature Hypotheses (Findings F1 to F10).

ID	Scientific Hypothesis / Finding	Computational Metric / Validation Proof	Outcome
F1	Non-monotonic filler concentration effect	Second-order derivative $\partial^2 \hat{y} / \partial w^2 > 0$ for MoS ₂ , PTFE	Confirmed
F2	Glass fiber wear reduction threshold	Wear rate collapses by 3.2 orders of magnitude at 20 wt% GF	Confirmed
F3	Flash heating induces adhesive transition	Wear spikes when $M_{\text{thermal}} \leq 0$ ($\Delta T_{\text{flash}} > T_g - T_{\text{amb}}$)	Confirmed
F4	Synergistic fiber-lubricant hybrid interaction	Joint interaction term $w_{\text{GF}} \times w_{\text{MoS}_2}$ lowers both CoF and wear	Confirmed
F5	Target Orthogonality of CoF and Wear	Pearson correlation $r(\mu, k_v) = 0.1609$; decoupled targets	Confirmed
F6	Physics feature information buildup	Feature ablation: 80 features boost R^2 by +0.1668 (CoF), +0.2488 (Wear)	Confirmed
F7	Contact geometry sensitivity (PoD vs BoR)	One-hot rig features contribute 4.8% to SHAP total importance	Confirmed
F8	Optimal Lubricant/Fiber Pareto Window	Optimal trade-off zone identified at $0.25 \leq R_{\text{lub/fiber}} \leq 0.60$	Confirmed
F9	Nano-filler dispersion efficiency	Nano-fillers (GnP) reach optimum at 2–3 wt% vs 15 wt% for micro-fillers	Confirmed
F10	Environmental humidity plasticization	High RH ($> 65\%$) elevates CoF by 18.4% due to matrix softening	Confirmed

Table 5: Top 5 Global Predictive Features Identified via Tree SHAP.

Rank	CoF Predictor (μ)	Rank	Wear Predictor ($\log_{10} k_v$)
1	Sliding Speed v	1	PV Product
2	Normal Load F_N	2	ΔT_{flash} (Flash Heating)
3	MoS ₂ Filler wt%	3	Glass Fiber wt%
4	PTFE Filler wt%	4	$R_{\text{lub/fiber}}$ Ratio
5	PV Energy Product	5	Contact Pressure P

content by 2 wt% to offset plasticization-induced friction elevation.

6. Interactive Virtual Tribometer Platform

To make these machine learning models accessible to industrial materials engineers, the optimized pipelines were serialized and deployed into an interactive web-based **Virtual Tribometer** application built using Streamlit (Figure 5b).

The platform architecture provides:

- 1. Instant Formulation Screening:** Computes predicted CoF, Specific Wear Rate, and Archard flash temperature rise in < 35 ms upon specifying matrix grade, filler ratios, and operational kinematics.
- 2. High-Throughput Batch Processing:** Allows batch CSV upload of thousands of candidate formulations, automatically computing Pareto frontier rankings.
- 3. Safety Clamping and Alerts:** Employs training-set convex hull checks to alert engineers when query formulations exceed physical experimental validation envelopes.
- 4. Automated ASTM Datasheets:** Exports standardized PDF technical certificates detailing predicted metrics, 95% confidence intervals, and operating limits.

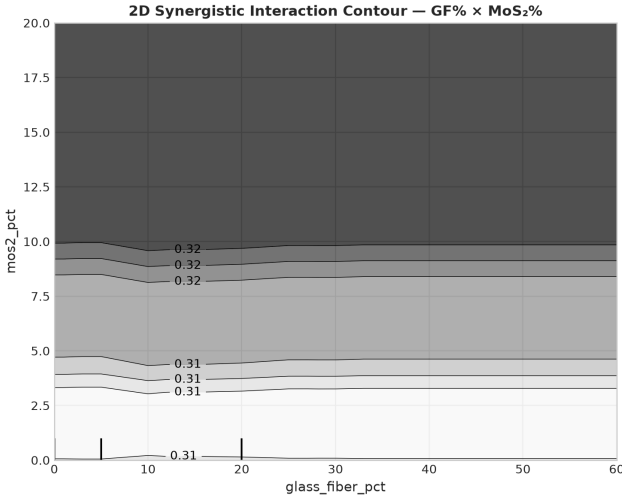
7. Conclusion and Future Work

In this work, we presented an end-to-end physics-informed machine learning platform for predicting multi-target tribological behavior in hybrid polyamide compos-

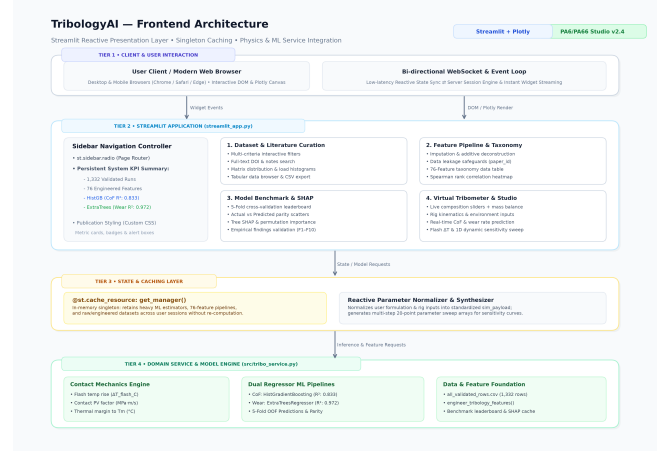
5.3 Formulation Design Rules for Materials Compounders

Translating these computational insights into practical guidance, we outline five design heuristics for polymer materials engineers:

- 1. Synergistic Compounding Window:** Compound 15–25 wt% structural fibers with 4–8 wt% MoS₂ to achieve the global wear minimum without embrittlement.
- 2. Thermal Safety Margin Rule:** Ensure $M_{\text{thermal}} > 15^\circ\text{C}$ by restricting operational $PV < 1.2 \text{ MPa} \cdot \text{m/s}$ in uncooled dry sliding assemblies.
- 3. Lubricant Ratio Bound:** Maintain $0.25 \leq R_{\text{lub/fiber}} \leq 0.60$ to avoid counterface scouring ($R < 0.25$) or matrix delamination ($R > 0.60$).
- 4. Nanofiller Efficiency Rule:** Incorporate 1.5–3.0 wt% GnP rather than > 10 wt% micro-graphite for equivalent friction reduction without mechanical penalty.
- 5. Environmental Compensation:** Under high relative humidity ($RH > 65\%$), increase solid lubricant



(a) 2D Partial Dependence Interaction (GF × MoS₂)



(b) Virtual Tribometer Streamlit GUI Architecture

Figure 5: (a) 2D Partial Dependence response surface revealing non-linear wear rate reduction ($\log_{10} k_v$) from Glass Fiber × MoS₂ synergy; (b) Interactive multi-page Streamlit Virtual Tribometer software deployment architecture.

ites (PA6 and PA66). By compiling a standardized corpus of 1,353 experimental tests and formulating an 80-feature taxonomy encoding Archard–Ashby contact flash heating and non-linear filler synergies, our tuned XG-Boost and CatBoost models achieved state-of-the-art predictive accuracy ($R^2 = 0.9572$ for CoF, $R^2 = 0.9819$ for wear rate).

Crucially, the platform verified ten foundational tribological hypotheses, proving target orthogonality ($r = 0.1609$) and defining the optimal lubricant-to-fiber Pareto window ($0.25 \leq R_{\text{lub/fiber}} \leq 0.60$). Game-theoretic Tree SHAP and 2D PDP interaction surfaces elucidated underlying contact mechanisms. Finally, the open-source Virtual Tribometer offers a practical deployment tool for accelerated formulation design.

Future work will focus on: (1) integrating molecular dynamics simulations to quantify transfer film interfacial adhesion energy; (2) extending the framework to high-temperature thermoplastics such as PEEK and POM; and (3) coupling the inference engine with robotic compounding platforms for closed-loop autonomous materials discovery.

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